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Effect of concentration on spectral properties of lanthanide ions-doped fluorophosphate glasses

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Abstract

Concentration dependent spectroscopic properties of Ln^{3+} ($\text{Ln} = \text{Nd}, \text{Sm}, \text{Dy}$ and Yb) -doped fluorophosphate glasses have been investigated. From the absorption spectra, Judd-Ofelt (JO) intensity parameters have been evaluated which are in turn used to predict radiative properties for the fluorescent levels of the said Ln^{3+} ions. For example, three JO intensity parameters (Ω_2 , Ω_4 and Ω_6) for the Dy^{3+} ions in fluorophosphate glass are found to be $\Omega_2 = 12.34 \times 10^{-20} \text{ cm}^2$, $\Omega_4 = 2.67 \times 10^{-20} \text{ cm}^2$ and $\Omega_6 = 2.30 \times 10^{-20} \text{ cm}^2$. Using FTIR spectra, OH^- content in 1.0 mol % Nd_2O_3 -doped samples, responsible for quenching the lifetimes of Nd^{3+} ions, is found to be $2.01 \times 10^{20} \text{ ions/cm}^3$ which is very much less compared to other reported glass systems. The decay profiles for the $^4\text{G}_{5/2}$ level of Sm^{3+} ions in the fluorophosphate glasses are found to change from single exponential (0.1 mol%) to non-exponential (0.5, 1.0, 2.0, 4.0 and 6.0 mol%) with shortening of lifetimes. The Yb^{3+} -doped fluorophosphate glasses possess high absorption (2.55 cm^2) and emission (3.44 cm^2) cross-sections and high figure of merit (3.33). The analysis of the results confirm that the Ln^{3+} -doped fluorophosphate glasses could be useful for the development of photonic devices.

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1. Introduction

Research on lanthanide ions (Ln^{3+})-doped optical quality transparent glasses are of great interest as they find applications in solid state lasers, optical amplifiers, upconverters, optical sensors, quantum cutters for solar energy harvesting, and phosphors [1-2]. The performance of these devices is limited by fast energy transfer, non-radiative de-excitation probabilities, clustering of Ln^{3+} ions at higher concentrations [3-4]. In order to improve their performance, it is necessary to limit the clustering of Ln ions. In this regard, the selection of chemical composition and concentration of Ln^{3+} ions plays vital role in the formation of clusters in glasses [5].

The optical properties of Ln^{3+} ions in glasses depend on the chemical composition of the glass, which determines the structure and nature of the bonds. Among different host materials, fluorophosphate (FPh) glasses are promising hosts because these glasses can combine the advantages of both fluoride and phosphate glasses such as lower phonon energy, better moisture resistance, physical and chemical stability, lower nonlinear refractive index and higher transparency from near UV to mid IR spectral range [6-8]. It was also found that a relatively higher degree of line broadening and smoother line shapes can be obtained [9] with FPh glass. These properties contribute to the applications of FPh glasses as Ln^{3+} -doped fiber laser. Many researchers have showed that Ln^{3+} -doped FPh glasses have promising features as laser gain media [10–12]. It is well known that the spectroscopic parameters such as absorption cross-section, emission cross-section and fluorescence lifetimes for metastable states of Ln^{3+} ions depend on their local environment in the glass matrix [13,14]. For FPh glasses, the local structure of Ln^{3+} ions can be modified over a large extent by varying P/F ratio and/or introducing other special ions [15,16].

2. Experimental details

Lanthanide (Ln^{3+})-doped transparent fluorophosphate glasses with molar compositions of $\text{P}_2\text{O}_5 + \text{K}_2\text{O} + \text{MF} + \text{SrO} + \text{Al}_2\text{O}_3 + \text{Ln}_2\text{O}_3$ (where $\text{MF} = \text{KF/MgF}_2/\text{SrF}_2/\text{AlF}_3$; $\text{Ln} = \text{Nd/Sm/Dy/Yb}$) ($x = 0.1, 0.5, 1.0, 2.0, 4.0$ and 6.0 mol %) were prepared by conventional melt quenching technique. The method of preparation and experimental techniques for spectroscopic measurements [17].

3. Results and discussion

3.1. Absorption spectra

From the absorption spectrum, the experimental oscillator strengths of various absorption transitions are evaluated. A least-square fitting procedure has been used to determine the JO parameters following the procedure described elsewhere [17]. As can be seen from Table 1, the trend of the JO parameters is $\Omega_2 > \Omega_6 > \Omega_4$ for PKMFAN10 glass and $\Omega_6 > \Omega_2 > \Omega_4$ for PKSFAN10 glass. The larger value of Ω_2 is due to the relatively higher value of the oscillator strength of the hypersensitive transition. The parameter Ω_2 is affected by the covalency of the Ln-O bond as well as the asymmetry around the Ln^{3+} ion site. In the present case, the higher value of Ω_2 indicates less centrosymmetric coordination [18] and a strong covalence [19] nature of metal-ligand bond. The condition $\Omega_6 \gg \Omega_4$ favours the intense $^4\text{F}_{3/2} \rightarrow ^4\text{I}_{11/2}$ transition at $1.06 \mu\text{m}$ and $\Omega_6 \ll \Omega_4$ favours the intense $^4\text{F}_{3/2} \rightarrow ^4\text{I}_{9/2}$ transition at $0.89 \mu\text{m}$ [20] compared to other transitions in the emission spectra. For present glass systems, the spectroscopic quality factor ($\chi = \Omega_4/\Omega_6$) values are 0.70 (PKMFAN10) and 0.74 (PKSFAN10) which are presented in Table 1. For the PKFSASm10 glass system, the trend is $\Omega_2 > \Omega_4 > \Omega_6$, the higher value of Ω_2 indicates that PKFSAS10 glass has stronger covalent nature while the other parameters, Ω_4 and Ω_6 are related to the rigidity of the host. It is observed that the $^4\text{G}_{5/2} \rightarrow ^6\text{H}_{9/2}$ transition exhibits higher $\sigma(\lambda_p)$ value for PKFSAS10 glass, which signifies the rate of energy extraction from the laser material. For the PKBAFDy10 glass, the larger value of Ω_2 indicates larger degree of covalency between the Dy^{3+} ion and the surrounding ligands and/or lower symmetry of the coordination structure surrounding the Dy^{3+} ion. As can be seen from Table1, for the PKBAFDy10 glass, the quantum efficiency exceeds 100%. This inconsistent result is attributed to the inherent uncertainty involved in the approximations of the JO theory.

Table 1. Refractive indices (n), concentrations (N_0 , 10^{20} ions/cc), Judd-Ofelt parameters (Ω_i , $\times 10^{-20}$ cm²), radiative (τ_R) and experimental (τ_{exp}) lifetimes (μ s), quantum efficiencies (η) and stimulated emission cross-sections ($\sigma(\lambda_p)$, $\times 10^{-22}$ cm²) for 1.0 mol% Ln₂O₃-doped fluorophosphate glasses.

S.No.	Glass	n	N_0	Ω_2	Ω_4	Ω_6	Trend	τ_R	τ_{exp}	η %	$\sigma(\lambda_p)$
1.	PKSFANd10	1.540	2.48	5.24	4.30	5.81	$\Omega_6 > \Omega_2 > \Omega_4$	326	211	65	3.29
2.	PKMFANd10	1.524	2.55	4.41	2.88	4.06	$\Omega_2 > \Omega_6 > \Omega_4$	491	200	41	1.81
3.	PKFSASm10	1.515	2.69	8.36	7.42	4.37	$\Omega_2 > \Omega_4 > \Omega_6$	1840	1773	96	14.12
4.	PKBAFDy10	1.539	2.46	12.34	2.67	2.30	$\Omega_2 > \Omega_4 > \Omega_6$	669	732	109	4.76

3.2. Decay curve analysis

Luminescence decay curves of the Sm³⁺ ions have been measured by exciting with 476.5 nm laser and monitoring the 645 nm emission. The concentration dependent decay profiles for the ⁴G_{5/2} level of Sm³⁺ ions in PKFSASm glasses are shown in Fig. 1. As can be seen, the decay curves change from single exponential to non-exponential with increase in Sm³⁺ ion concentration. Hence, the experimental lifetimes (τ_{exp}) of the ⁴G_{5/2} level in PKFSASm glasses have been calculated as per the method described in Ref [17]. The τ_{exp} values of the ⁴G_{5/2} level are found to be 2699, 1863, 1773, 1177, 331 and 108 μ s for 0.1, 0.5, 1.0, 2.0, 4.0, 6.0 mol% Sm₂O₃-doped PKFSA glasses, respectively. The observed non-exponential nature with shortening of lifetimes could be explained by the presence of an excitation energy transfer and cross-relaxation channels: (⁴G_{5/2}, ⁶H_{5/2}) → (⁶F_{5/2}, ⁶F_{11/2}) and (⁴G_{5/2}, ⁶H_{5/2}) → (⁶F_{9/2}, ⁶F_{7/2}) from Sm³⁺ ion being in the excited ⁴G_{5/2} state to another one in the ground state. Later the two ions relax non-radiatively to their ground state, ⁶H_{5/2}. This transfer can occur either by exchange coupling or by electric multipolar interactions.

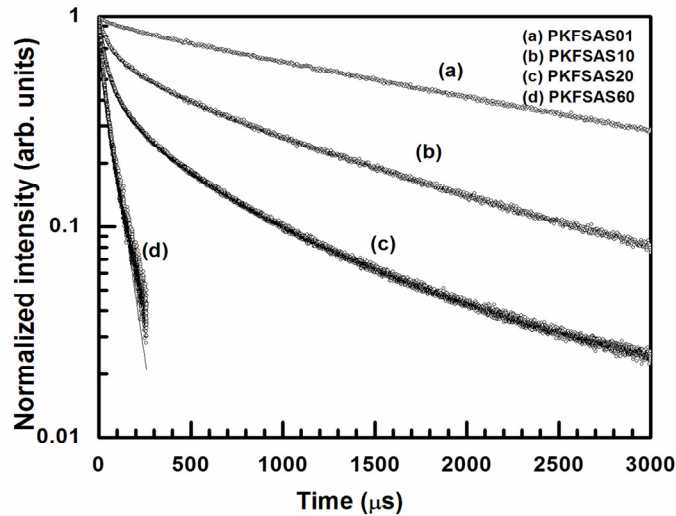


Fig.1. Decay curves of the ⁴G_{5/2} level in (a) PKFSAS01 (b) PKFSAS10 (c) PKFSAS20 and (d) PKFSAS60 glasses.

3.3. FTIR spectra

The presence of OH⁻ impurities in the studied glasses plays a key role in photonic device performance. In order to know the OH⁻ concentrations in the glasses, the FTIR spectra are measured for 1.0 mol% Nd₂O₃-doped glasses and are shown in Fig. 2.

From the FTIR spectra, OH⁻ concentrations in 1.0 mol% Nd₂O₃-doped glasses, responsible for quenching the lifetimes of Nd³⁺ ions, are calculated to be 2.01×10^{20} and 0.98×10^{20} ions/cm³ in PKMFAN10 and PKSFAN10 glasses, respectively, which are very much less than 5.82×10^{20} ions/cm³ reported for 0.1 mol% Er₂O₃-doped zinc tellurite glass [21]. From this analysis, it is clear that PKSFAN10 glass is having lower OH⁻ content resulting in smaller

quenching effect on Nd^{3+} ions fluorescence in this glass. This is one of the reasons for the higher lifetimes of the $^4\text{F}_{3/2}$ level in PKSFAN glasses compared to PKMFAN glasses for all the three Nd^{3+} ion concentrations studied in the present work (Table 1).

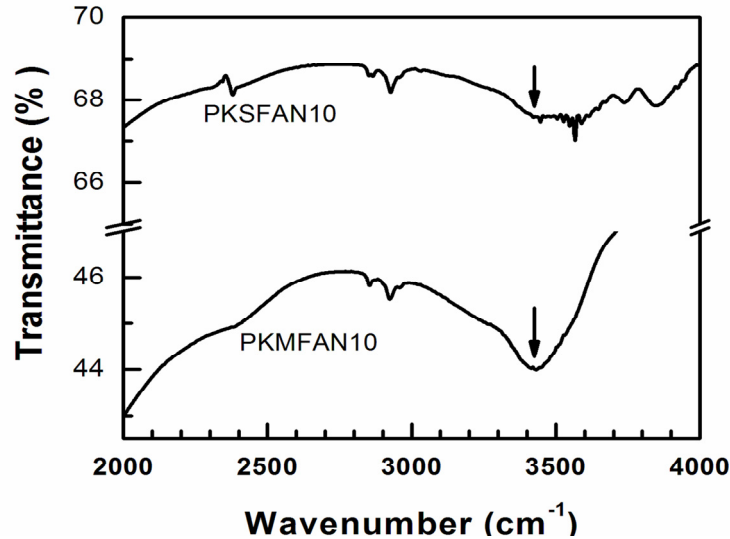


Fig. 2. The FTIR spectra of PKMFAN10 and PKSFAN10 glasses.

3.4. Laser performance parameters

In order to assess the potential of PKFSAYb glasses for laser devices, the laser performance parameters such as absorption (σ_{ab}) and emission (σ_{em}) cross-sections, lifetimes (τ), β_{min} (minimum fraction of Yb^{3+} ions that must be excited to balance the gain exactly with the ground state absorption at a laser wavelength), I_{sat} (pump saturation intensity) and I_{min} (minimum absorbed pumping intensity) have been determined following the procedure described elsewhere [22] and are shown in Table 2. As can be seen from Table 2, radiative lifetime (τ_{R}) increases with increase in Yb_2O_3 concentration in PKFSAYb glasses. The values of β_{min} , I_{sat} and I_{min} are in the range of 0.133 - 0.463, 9.52 - 100.4 kW/cm^2 , 2.46 - 15.57 kW/cm^2 , respectively. There is a considerable change in the value of β_{min} for the PKFSAYb glasses. It is desirable for I_{sat} to be as low as possible to minimize the minimum pumping intensity (I_{min}) as the emission cross-section is proportional to absorption cross-section. For a good laser glass, figure of merit ($\sigma_{\text{em}} \times \tau_{\text{exp}}$) and energy extraction efficiency are generally desirable to be as large as possible to provide high gain and the I_{min} to be as small as possible to minimize the pump losses. From Table 2, it is noticed that PKFSAYb01 glasses possess higher figure of merit value of 3.33, compared to reported glasses. Hence, PKFSAYb01 glass is more suitable for laser applications.

Table 2. Spectroscopic and laser performance parameters of Yb^{3+} -doped fluorophosphate glasses.

S. No.	Glass label	n	σ_{ab} (cm^2)	σ_{em} (cm^2)	τ_{R} (ms)	τ_{exp} (ms)	β_{min}	I_{sat} (kW/cm^2)	I_{min} (kW/cm^2)	$\sigma_{\text{em}} \times \tau_{\text{exp}}$	SFL
1.	PKFSAYb01	1.515	2.55	3.47	1.02	0.96	0.463	9.52	4.41	3.33	0.755
2.	PKFSAYb05	1.516	1.74	2.31	1.14	0.86	0.159	15.51	2.46	1.99	0.809
3.	PKFSAYb10	1.517	1.63	2.18	1.19	0.64	0.142	22.27	3.18	1.39	0.440
4.	PKFSAYb20	1.517	1.61	2.16	1.30	0.47	0.133	30.54	4.07	1.01	0.251
5.	PKFSAYb40	1.526	1.29	1.72	1.53	0.29	0.140	62.51	8.75	0.50	0.057
6.	PKFSAYb60	1.529	1.10	1.48	1.46	0.21	0.155	100.40	15.57	0.31	0.020

4. Conclusions

The fluorophosphate glasses doped with various Ln^{3+} ions have been prepared by conventional melt-quenching technique and studied their concentration dependent spectroscopic and laser performance parameters. The spectroscopic quality factor for the present PKMFAN10 and PKSFA10 glasses indicate that there is more probability for the ${}^4\text{F}_{3/2} \rightarrow {}^4\text{I}_{11/2}$ emission transition at around 1.06 μm . The larger value of Ω_2 indicates the larger asymmetry at the Sm^{3+} ion site as well as strong covalent nature of Sm-O bond in the PKFSAS10 glass. It is observed that the ${}^4\text{G}_{5/2} \rightarrow {}^6\text{H}_{9/2}$ transition possess higher emission cross-section ($\sigma(\lambda_p)$). The radiative properties derived from the JO theory as well as concentration dependent lifetimes of the ${}^4\text{F}_{9/2}$ level have been reported. It is found that the PKFSAYb01 glass possess higher absorption and emission cross-sections and higher figure of merit. The measured lifetime of ${}^2\text{F}_{5/2}$ level decreases from 0.96 to 0.21 ms in fluorophosphate glasses when Yb_2O_3 concentration is increased from 0.1 to 6.0 mol% due to energy transfer between Yb^{3+} ions and OH^- ions.

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References

- [1] C. Görller-Walrand, K. Binnemans, in: K. A. Gschneidner Jr., L. Eyring (Eds.), Handbook on the Physics and Chemistry of Rare Earths, Vol.25, North Holland, Amsterdam, 1998 (ch.167).
- [2] X. Zou and H. Toratani, Phys. Rev. B 52 (1995) 15889-15897.
- [3] B. J. Ainslie, S. P. Craig-Ryan, S. T. Davey, J. R. Armitage, C. G. Atkins, J. F. Massicot and R. Wyatt, IEEE Proc. 137 (1990) 205-208.
- [4] P. Blixt, J. Nilsson, T. Carlnas and B. Jaskorzynska, IEEE Photon. Technol. Lett. 3 (1991) 996-998.
- [5] K. Arai, H. Namikawa, K. Kumata, T. Honda, Y. Ishii and T. Handa, J. Appl. Phys. 59 (1986) 3430-3436.
- [6] S.V.J. Lakshman, Y.C. Ratnakaram, Phys. Chem. Glasses 31(1990) 42-44.
- [7] B. Viana, M. Palazzi, O. Lefol, J. Non-Cryst. Solids 215 (1997) 96-102.
- [8] S. Jiang, T. Luo, B.C. Hwang, F. Smekatala, K. Seneschal, J. Lucas, N. Peyghambarian, J. Non-Cryst. Solids 263-264 (2000) 364-368.
- [9] D. Ehrt, Proc. SPIE 1761 (1992) 213-222
- [10] K. Lu, N.K. Dutta, J. Appl. Phys. 91 (2002) 576-581.
- [11] L. Zhang, H.F. Hu, J. Non-Cryst. Solids 292 (2001) 108-114.
- [12] H.B. Yin, P.Z. Deng, J.Z. Zhang, H.B. Yin, P.Z. Deng, J.Z. Zhang, J. Non-Cryst. Solids 210 (1997) 243-248.
- [13] G.Y. Zhang, M.J. Poulain, G.Y. Zhang, M.J. Poulain, J. Alloys Compd. 275-277 (1998) 15-20.
- [14] B. Karmakar, P. Kundu, R.N. Dwivedi, J. Non-Cryst. Solids 289 (2001) 155-162.
- [15] D. Ehrt, W. Wintzer, W. Topfer, Proceedings of the XIII International Symposium on Non-oxide Glasses and New Optical Glasses, Pardubice, Czech Republic, 2002, p. 662.
- [16] T. Murata, K. Mazeno, K. Morinaga, Sci. Technol. Adv. Mater. 3 (2002) 85-90.
- [17] R. Vijaya, V. Venkatramu, P. Babu, C.K. Jayasankar, U. R. Rodríguez-Mendoza, V. Lavín, J. Non-Cryst. Solids 365 (2013) 85-92.
- [18] Z. Jiang, J. Yang and S. Dai, J. Opt. Soc. Am. B 21 (2004) 739-743.
- [19] C.K. Jorgensen and R. Reisfeld, J. Less-Common Met. 93 (1983) 107-112.
- [25] R. C. Powell, Physics of Solid State Laser Materials, Springer, New York, 1998.
- [21] S. Dai, J. Zhang, C. Yu, G. Zhou, G. Wang and L. Hu, Mater. Lett. 59 (2005) 2333-2336.
- [22] L.D. Loach, S.A. Payne, L.K. Smith, W.L. Kway, W.F. Krupke, J. Opt. Soc. Am. B:Opt. Phys. 11 (1994) 269-276.